

Red Team AI Malware Simulation Report

Target: localhost

Generated: 2025-10-26 12:42 UTC

Simulation: 2025-10-26 12:42 → 2025-10-26 12:42

Version: Red Team AI Engine v2025.09.11

Author / Team: Red Team / Security Assessment Team

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Executive Summary

Purpose of Simulation:

This simulation assessed the security posture of the target system and tested its defenses against a controlled AI-driven malware attack. The goal was to evaluate how the system responds to various automated offensive techniques, including evasion, reconnaissance, and attack decision-making.

Key Findings:

- **Open Ports / Services of Concern:**

- 21 - Service detected
- 80 - Service detected
- 631 - Service detected

- **Sensitive Files or Data Identified:**

- None

- **Evasion Success Rate:** 85%

- **Active AV / EDR Detected:**

- ClamAV

- **Overall Risk Assessment:** Low

AI Model Performance:

- **Evasion AI:** 85% — Successfully evaded detection for 85% of actions.
- **Recon Prioritisation AI:** 20% — Correctly prioritized 20% of reconnaissance findings.
- **Attack Decision AI:** 60% — Selected optimal attack actions 60% of the time.

Scope & Methodology

Overview

This engagement is a controlled, simulated assessment executed by the RAPTOR Red Team AI Malware Simulator. The purpose is to evaluate detection, prioritisation, and response capabilities of the target system in a safe research environment — not to perform real-world exploitation. All activity is constrained to authorized test systems and simulated payloads.

Phases

Phase 1 — Reconnaissance (Passive & Active Collection)

Objective: gather an inventory of the target system (OS, services, users, exposed ports, environment variables, and candidate sensitive files) to produce prioritized reconnaissance findings for later phases. Scope: passive discovery first (public-facing services, metadata), followed by limited, consented active scanning in a lab-like environment. Deliverables: structured recon dataset, open port list, environment snapshot, and prioritized asset list.

Phase 2 — C2 Simulation Dummy Actions (Controlled Emulation)

Objective: emulate attacker post-compromise behaviour at a high level — command-and-control interaction, staged dummy actions, and controlled data movement — while avoiding any real malicious payloads or persistence mechanisms. Scope: simulated C2 exchanges and benign stand-in payloads that exercise detection logic and telemetry generation only. Actions are designed to provoke detection and measure evasion, not to exploit vulnerabilities.

Phase 3 — Reporting & Analysis

Objective: consolidate data from prior phases into an evidence-backed report, quantify model performance, and provide mitigations and recommendations. Deliverables: structured findings, charts (e.g., evasion success), per-module analysis, and an executive summary suitable for stakeholders.

AI Models and Roles

Evasion AI (Payload-side, Learns Detector Signals)

Role: adapt simulated offensive behaviour to reduce detection signals observed in the sandboxed environment. The model optimizes for increased “stealth” as measured by telemetry/sensor responses and detection flags. Outputs: an evasion score, action-level success rates, and a ranked list of actions that consistently avoid detection. Notes: this model operates solely against simulated detectors and logged telemetry; it does not create

or deliver real-world exploits.

Recon Prioritisation AI (Server-side, Post-processing)

Role: ingest raw reconnaissance artifacts (service lists, file discovery, logs) and rank findings by operational relevance and risk. It helps the analyst focus on high-value assets and probable attack vectors. Outputs: prioritized recon list, confidence scores, and suggested follow-ups.

Attack Decision AI (Server-side, Planner)

Role: select the next benign/simulated action based on recon priorities, evasion feedback, and mission objectives. It balances likelihood of success, detection risk, and mission value to recommend the best next simulated step. Outputs: ordered action plans and rationale metadata (e.g., why an action was chosen).

Data, Metrics & Evaluation

Key metrics captured during simulations include:

- **Evasion Success Rate:** percentage of simulated actions that did not trigger detection flags.
- **Detection Events:** timestamped telemetry and alert counts for each simulated action.
- **Recon Prioritisation Accuracy:** analyst-validated relevance of top-N findings.
- **Decision Optimality:** proportion of selected actions that met mission objectives under risk constraints.

Evaluation approach: metrics are aggregated per-simulation and across runs; charts (e.g., pie/bar) visualise distributions. All model outputs are accompanied by confidence values and human-readable explanations to aid analyst review.

Safety, Ethics & Operational Controls

This simulation adheres to strict safety and ethical controls:

- **Authorization:** tests are executed only on systems explicitly authorized for simulation. No unsanctioned scanning or exploitation is performed.
- **No Real Exploits:** payloads are inert or simulated placeholders; persistence, credential theft, or destructive actions are not performed.
- **Logging & Isolation:** all activity is logged, contained within isolated test networks, and retained for post-test analysis.
- **Human Oversight:** model decisions and high-risk actions require human review before enactment in any semi-automated workflow.

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- **Compliance:** users of the software must comply with applicable laws, institutional policies, and the RAPTOR license terms.

Tools, Techniques & Limitations

Additional Techniques: Automated reconnaissance, AI-based file sensitivity analysis, intelligent attack planning

Limitations: results are representative for the test environment and detectors used; real production systems and defensive stacks may behave differently. Model performance depends on the quality of telemetry and labelled data available.

Recon Findings

Host Information

- Operating System: Linux 85 22.04.1-Ubuntu SMP PREEMPT_DY $\textit{N}$ AMICFriSep1916 : 18 : 59UTC2Architecture : 64bit
- Patch Level / Updates: N/A
- Installed Software:
 - N/A

User & Account Details

- Privileged Accounts:
 - None
- Active Users:
 - None
- Default Credentials Detected: None

Network Topology

- IP Addresses: N/A
- Subnets: N/A
- Firewalls: None
- VPN / DMZ / Segmentation Info: N/A

Open Ports & Services

- Port 21 / tcp — Service: unknown vN/A
- Port 80 / tcp — Service: unknown vN/A
- Port 631 / tcp — Service: unknown vN/A

Active Processes & Services

- N/A

Endpoints & Devices

- None

Active AV / EDR

- None detected

Misconfigurations Detected

- None

Findings & Analysis

This section analyzes the data collected by the Red Team AI Malware Simulator, highlights critical issues, and provides context for the AI’s actions and attack outcomes. It focuses on risk assessment and patterns identified during the simulation.

Finding / Issue	Evidence / Data	Severity	Potential Impact if System Were Real
Sensitive Environment Variables	1 environment variables with potentially sensitive names detected	Medium	Possible credential or API key exposure

Evasion Metrics

Purpose: This section measures the effectiveness of the Evasion AI in avoiding detection by simulated defenses. It is objective and metric-focused — no interpretation or recommendations are provided here.

Detection mechanisms tested:

- Signature-based AV
- Host telemetry
- Network IDS

AV / EDR presence (detected on system):

- None detected

VM / Sandbox detection indicators:

- No VM/sandbox detection heuristics observed

Monitoring / Logging mechanisms observed:

- None / N/A

Evasion success summary:

Overall Success Rate: **85%** (The payload avoided detection in 85% of recorded checks.)

Success by defense type:

Defense Type	Success Rate (%)
No per-defense breakdown available	N/A

Actions skipped or delayed by Evasion AI:

- None

Attack Simulation Summary

Purpose: This section summarizes the actions executed by the simulated malware, including their outcomes and prioritization. It provides an overview of what was attempted, what succeeded or failed, and how the AI prioritized each action.

Actions Executed: The table below lists all simulated attack actions performed by the Red Team AI Malware Simulator. Each action includes a description, the outcome, and its priority or importance score as determined by the Recon Prioritization AI.

Action Name	Description	Outcome	Priority / Score
<i>file_enumeration</i>	AI recommended action based on analysis (confidence: 60.00		

LLM Based Mitigations

Purpose: This section uses a pre-trained Large Language Model (LLM) to analyze the collected simulation data and generate recommended mitigations for the findings. It provides a concise “briefing for action” rather than a deep technical manual.

Note on LLM Limitations:

- Recommendations are based solely on the simulation data provided.
- The LLM may not capture every nuance of real-world system configurations or risks.
- Human verification and contextual analysis are recommended before applying any mitigations.

Recommended Mitigations:

- Implement principle of least privilege - avoid running applications with admin rights
- Deploy endpoint detection and response (EDR) solution
- Enable comprehensive logging and monitoring
- Regular security awareness training for users
- Conduct periodic vulnerability assessments and penetration tests

Raw Data

```
{
  "target_name": "localhost",
  "generated_at": "2025-10-26T12:42:10Z",
  "sim_start": "2025-10-26 12:42",
  "sim_end": "2025-10-26 12:42",
  "exec_summary": {
    "purpose": "AI-driven penetration test simulation for security assessment",
    "open_ports_list": [
      "21 - Service detected",
      "80 - Service detected",
      "631 - Service detected"
    ],
    "sensitive_data_list": [],
    "evasion_success": 85,
    "av_list": [
      "ClamAV"
    ],
    "overall_risk": "Low",
    "evasion_ai": 85,
    "recon_ai": 20,
    "attack_ai": 60
  },
  "scope": {
    "techniques": "Automated reconnaissance, AI-based file sensitivity analysis, inte",
    "phases": [
      "Phase 1 - System Reconnaissance",
      "Phase 2 - AI-Powered Threat Analysis",
      "Phase 3 - Attack Simulation & Reporting"
    ],
    "ai_models": [
      {
        "name": "Recon Prioritization AI",
        "role": "Identifies sensitive files using LightGBM classifier (95% accuracy)"
      },
      {
        "name": "Attack Decision AI",
        "role": "Predicts optimal attack sequence based on environment state"
      }
    ]
  },
  "recon_data": {
```

```
"os_name": "Linux",
"os_version": "#85~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Fri Sep 19 16:18:59 UTC 2",
"os_release": "6.8.0-85-generic",
"architecture": "64bit",
"hostname": "localhost",
"current_user": "prasdud",
"machine": "x86_64",
"processor": "x86_64",
"is_admin": false,
"open_ports": [
  {
    "number": 21,
    "protocol": "tcp",
    "service_name": "unknown",
    "version": "N/A",
    "vulns": []
  },
  {
    "number": 80,
    "protocol": "tcp",
    "service_name": "unknown",
    "version": "N/A",
    "vulns": []
  },
  {
    "number": 631,
    "protocol": "tcp",
    "service_name": "unknown",
    "version": "N/A",
    "vulns": []
  }
],
"env_vars_count": 62,
"installed_software": [],
"processes": []
},
"findings": [
  {
    "name": "Sensitive Environment Variables",
    "evidence": "1 environment variables with potentially sensitive names detected",
    "severity": "Medium",
    "impact": "Possible credential or API key exposure"
  }
],
```

```
"evasion": {
  "detection_mechanisms": [
    "Signature-based AV",
    "Host telemetry",
    "Network IDS"
  ],
  "overall_success": 85,
  "skipped": []
},
"attacks": [
  {
    "name": "file_enumeration",
    "description": "AI recommended action based on analysis (confidence: 60.00%)",
    "outcome": "Simulated",
    "priority": 6
  }
],
"mitigations": [
  "Implement principle of least privilege - avoid running applications with admin r",
  "Deploy endpoint detection and response (EDR) solution",
  "Enable comprehensive logging and monitoring",
  "Regular security awareness training for users",
  "Conduct periodic vulnerability assessments and penetration tests"
]
}
```